PREDICTABLE DESIGNS

Guide: Selecting the Best STM32 Microcontroller for Your Project

STM32 Series:

F series	Original series. 90nm process
L series	Low-power series. 90nm process
U series	Low-power performance. Only U5 right now. 40nm process
H series	Highest performance series. 40nm process
G series	Budget series. 90nm process
C series	Ultra low-cost series. Only C0 right now.
W series	Wireless series- BLE, Zigbee, Thread, and long-range version with LoRaWan
MP series	Microprocessors (Not covered in this video)

Before digging into specifics for each series, it's important to understand the differences between the various ARM Cortex-M core versions which all STM32's use.

Cortex-M variations:

Cortex-M0	Most basic, low cost
Cortex-M0+	Most energy efficient, smallest core, lowest cost, Memory Protection Unit (MPU)
Cortex-M3	Mid-point between cost and performance. High-performance, real-time processing for cost-constrained applications.
Cortex-M4	Adds dedicated DSP, optional single-precision FPU, higher max clock speed, same DMIPS/MHz as M3
Cortex-M4F	Commonly (but not always used) to signify an M4 with an optional FPU
Cortex-M7	2x DMIPS/MHZ as M3/M4, optional double-precision FPU, Higher max clock speed
Cortex-M33	Enhanced security features, FPU, DSP extension, 20% greater performance than M4. Uses ARMv8 instruction set, versus all other M cores use ARMv7.

Grouped by processor core(s):

STM32F0	Cortex-M0 @ 48MHz
STM32C0	Cortex-M0+ @ 48MHz
STM32G0	Cortex-M0+ @ 64MHz
STM32L0	Cortex-M0+ @ 32MHz
STM32F1	Cortex-M3 @ 72MHz
STM32F2	Cortex-M3 @ 120MHz
STM32L1	Cortex-M3 @ 32MHz
STM32F3	Cortex-M4F @ 72MHz
STM32F4	Cortex-M4F @ 180MHz
STM32G4	Cortex-M4F @ 170MHz
STM32L4	Cortex-M4F @ 80MHz
STM32L4+	Cortex-M4F @ 120MHz
STM32H5	Cortex-M33 @ 250MHz
STM32L5	Cortex-M33 @ 110MHz
STM32U5	Cortex-M33 @ 160MHz
STM32F7	Cortex-M7F @ 216MHz
STM32H7	Cortex-M7F @ 600MHz + Cortex-M4 @ 240MHz
STM32WL	Cortex-M4 @ 48MHz + Cortex-M0+ @ 48MHz -Long-range LoRaWAN
STM32WB	Cortex-M4 @ 64MHz + Cortex-M0+ @ 32MHz - BLE5.4 + ZigBee + Thread
STM32WBA	Cortex-M33 @ 100MHz - BLE5.4 + ZigBee + Thread
STM32WB0	Cortex-M0+ @ 64MHz - BLE5.3

Priority - Performance:

F and H series are highest performance

- F2 or F3 for moderate performance
- F4 for high performance
- F7, or H5 for very high performance
- H7 for highest level of performance 40nm process allows highest clock speeds

Priority - Power consumption:

L and U series (U5) have lowest power consumption with most low-power modes

The L-series is the primary low-power series.

The U5 is a new low-power option that uses their 40nm process instead of the 90nm process which allows it to have higher performance with higher clock speed than the L-series.

Most of the L-series is based on a Cortex-M4 core, but the L5 and U5 both instead use the Cortex-M33 core.

Priority - Security:

Select model based on Cortex-M33 core which include enhanced security features with an security extension called TrustZone.

STM32L5, STM32U5, or STM32H5 all based on Cortex-M33
STM32WBA based on M33 with BLE, Zigbee, Thread

Priority- Wireless:

STM32WL	LoRaWAN
STM32WB	BLE5.4 / ZigBee / Thread
STM32WBA	BLE5.4 / ZigBee / Thread
STM32WB0	BLE5.3 - Coming soon

Priority - By Advanced Features:

Camera interface	F4, F7 or H7
LCD-TFT display controller	F4, F7 or H7
MIPI-DSI display controller	A few select models of F4 and F7
Graphics accelerator	F4, F7, H7
Single-precision FPU	F4, F7, H5, H7
Double-precision FPU	F7 and H7
External RAM	

Priority - Cost:

C0, G series are optimized for lowest cost. F0 and F1 are also lower cost.

The C0 is the cheapest of all STM32's and it's designed to replace simple, low-cost 8/16-bit microcontrollers.

The G-series is the next cheapest group of STM32 microcontrollers.

Grouped by processor series with lowest Digikey cost:

STM32F0	Cortex-M0 @ 48MHz	$0.71 @ 5,000
STM32F1	Cortex-M3 @ 72MHz	$2.03 @ 500
STM32F2	Cortex-M3 @ 120MHz	$5.38 @ 1,000
STM32F3	Cortex-M4F @ 72MHz	$2.09 @ 3,000
STM32F4	Cortex-M4F @ 180MHz	$2.13 @ 2,500
STM32F7	Cortex-M7F @ 216MHz	$3.44 @ 2,400
STM32L0	Cortex-M0+ @ 32MHz	$0.68 @ 5,032
STM32L1	Cortex-M3 @ 32MHz	$2.31 @ 2,400
STM32L4	Cortex-M4F @ 80MHz	$1.83 @ 2,940
STM32L4+	Cortex-M4F @ 120MHz	$8.03 @ 80
STM32L5	Cortex-M33 @ 110MHz	$4.71 @ 1,040
STM32U5	Cortex-M33 @ 160MHz	$3.60 @ 2,500
STM32G0	Cortex-M0+ @ 64MHz	$0.63 @ 5,000
STM32G4	Cortex-M4F @ 170MHz	$2.58 @ 3,000
STM32C0	Cortex-M0+ @ 48MHz	$0.38 @ 4900
STM32H5	Cortex-M33 @ 250MHz	$1.92 @ 4,960
STM32H7	Cortex-M7F @ 600MHz + Optional Cortex-M4 @ 240MHz	$4.90 @ 990
STM32WL	Cortex-M4 @ 48MHz + Cortex-M0+ @ 48MHz - LoRaWAN	$3.59 @ 1,560
STM32WB	Cortex-M4 @ 64MHz + Cortex-M0+ @ 32MHz - BLE5.4/ZigBee/Thread	$1.91 @ 4,940
STM32WBA	Cortex-M33 @ 100MHz - BLE5.4 + ZigBee + Thread	$2.79 @ 2,400
STM32WB0	Cortex-M0+ @ 64MHz - BLE5.3	$1.36

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Meet your guide: John Teel

Hey there, I'm a former microchip design engineer at Texas Instruments and founder of a hardware startup that sold products in hundreds of retail stores. My chip designs are in devices from Apple, Intel, and more.

Now, my full-time focus is helping people like you bring new electronic products to life, without wasting time, money, or risking everything.

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